

Question 2: News Classification with TF-IDF Weighted TextCNN

Overview

You are given English-language news from AG News and must classify each article into one of 4 categories: World, Sports, Business, and Sci/Tech. The raw data contains URLs, numbers, backslashes, irregular whitespace, and some duplicate articles.

Clean the data, build TF-IDF features, and build a TextCNN with PyTorch. You may use scikit-learn for splitting the data, building TF-IDF features, and evaluation.

Description

The data provided is raw AG News data that has not been cleaned, has no duplicates removed, and comes with no separate validation set. You must clean the text (remove URLs, numbers, backslashes, unnecessary whitespace), remove duplicates, and split `train.csv` into your own train and validation portions before building TF-IDF and training the model. TF-IDF must be fit only on the text of the train portion after splitting.

- `train.csv` : 120,000 raw articles, with columns `id` , `text` , `label`
- `test.csv` : 7,600 raw articles, with only columns `id` , `text` , no label

The `label` column uses 0 = World, 1 = Sports, 2 = Business, and 3 = Sci/Tech.

Public Data

File structure under `data` :

```
train.csv
test.csv
```

`train.csv` has columns `id` , `text` , `label` , while `test.csv` has only `id` , `text` . The row order of `test.csv` matches the `id` used in the submission.

Evaluation

Evaluated with Accuracy and Macro-F1, where Macro-F1 weights each class equally.

```
Macro-F1 = mean(F1_World, F1_Sports, F1_Business, F1_SciTech)
```

Submission File

Submit `submission_nlp.csv` with 7,600 rows.

```
id,label  
17,2  
483,0  
921,3
```

- `id` : the id from `test.csv`
- `label` : the predicted class; must be 0, 1, 2, or 3.

Leagues of Code



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